Composition Shift and the Measurement of Bias in a Growing Prediction Market: Evidence from Kalshi, 2021–2026

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Abstract

Prediction markets are now studied intensively enough that their own growth has become a measurement problem. Using 376.8 million Kalshi fills through June 2026, we show that sports contracts went from 0.009% of the in-scope sample in calendar 2024 to 58.8% after May 2025, and that 41% of the aggregate change in the Mincer–Zarnowitz slope at that boundary is composition rather than any change in how the market prices risk. The natural before/after design across early 2025 largely measures the arrival of an asset class.

We then ask the persistence question properly. Reconstructing Kalshi's fee schedule from sixteen dated captures of its own published document, we find the 2025 maker fee was not the exchange-wide regime change the literature assumes: a per-series surcharge covering 8.0% to 14.4% of in-scope markets, revised five times, flat per contract before it was quadratic in price. That rules out a step dummy and calls for a difference-in-differences between treated and untreated series trading in the same months.

The pre-specified estimate is a tight null: -0.0016 (0.0064) on 98 treated series. An event study added after that result was known supports parallel trends ($\chi^2(5) = 3.24$, $p = 0.66$) and, read as exploratory, indicates a transitory reduction three to four months after a series is charged. The maker's return advantage at prices at or above 50c survives every plausible fee rate. We publish the fee history, the category mapping and the full pipeline.

Keywords: prediction markets, favorite–longshot bias, composition effects,

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Pre-specification

The R2 specification — the composition decomposition, the static difference-in-differences, the verdict thresholds — was committed on 2026-07-28, before any R2 estimate existed. **The event study in Section 6.3 was not**; it was added on 2026-08-27, after the static estimate was known. Section 6.4 states which of its two halves is a validity check and which is exploratory, and the abstract labels the exploratory result as such.

1. Introduction

The favorite–longshot bias — cheap contracts winning less often than their prices imply, expensive ones more often — is among the most replicated findings in the study of betting and prediction markets (Thaler and Ziemba, 1988; Ottaviani and Sørensen, 2008; Snowberg and Wolfers, 2010). Kalshi, the first federally licensed prediction market in the United States, has recently made it newly measurable at scale: transaction-level data are public, the venue is large, and its contracts settle unambiguously.

Two features of 2025 make it look like a natural experiment. Kalshi began charging fees to liquidity providers, taxing the one margin recent work identifies as exploitable. And in September 2025 the biases themselves were publicly documented in a widely circulated working paper, whose authors closed by inviting exactly the follow-up we conduct: whether the patterns persist "now that they have been publicly documented."

The natural design is a before/after comparison of the aggregate bias. **Our first result is that this design is contaminated, and we measure by how much.**

Kalshi did not merely grow across that boundary; it changed shape. Sports contracts launched in early 2025 — almost exactly at the boundary any such study would use — and by June 2026 they are 58.8% of the in-scope sample against 0.009% in calendar 2024. Because the bias is heterogeneous across categories, an aggregate comparison mixes a change in pricing behaviour with a change in what is being priced. Decomposed, 41% of the aggregate movement at the fee boundary is the composition term.

This is not a hypothetical concern. Kalshi is being analysed now, on samples that straddle the same boundary, and a naive aggregate comparison will report a large apparent effect.

Our remaining contributions follow from taking the confound seriously.

The fee treatment is not what it is assumed to be.. We reconstruct Kalshi's fee schedule from sixteen dated captures of the exchange's own published document, 2021-07 through 2026-02, plus per-series fee metadata for all 12,231 series. The maker fee was never exchange-wide: it is a surcharge on an enumerated list of series, 29 at introduction and 111 by September 2025, covering 8.0% to 14.4% of in-scope markets and 61% to 66% of volume. Its first form was flat per contract, becoming quadratic in price only on 2025-07-08. Two further carve-outs — a 0.14 taker rate before August 2021, and a half rate for index markets covering 18.3% of the pre-2025 universe — are not modelled in the existing literature at all.

Properly identified, the fee produced no persistent change in the bias.. Because treatment is per-series, staggered and revised, treated and untreated series trade in the same months and a difference-in-differences is available. The pre-specified estimate is -0.0016 (s.e. 0.0064) against never-treated series. Treatment here was selected on market-making activity, which is adjacent to the outcome, so the identifying assumption needs evidence rather than assertion; an event study added after that estimate was known finds pre-treatment coefficients jointly zero ($p = 0.66$), which supports it.

That same event study also finds post-treatment coefficients jointly non-zero ($p = 0.021$), suggesting a transitory reduction peaking three to four months after a series is charged. **We report this as exploratory**, for the reason Section 6.4 sets out: it is a hypothesis test run after a pre-specified test returned zero, which is the shape specification search produces even when nobody went looking. The naive step-dummy coefficient disagrees with both, for a third reason again: it is dominated by one category that has shrunk by a factor of fifty while retaining nearly a third of the composition-held-fixed weight.

The exploitable margin survives.. The maker's return advantage at prices at or above 50c is +2.40% gross in the post-boundary window and never crosses zero across the plausible band of maker fee rates.

Section 2 describes the data. Section 3 validates the pipeline against published results. Section 4 is the composition result. Section 5 is the fee schedule. Section 6 is the persistence estimate. Sections 7 and 8 cover robustness and related work.

2. Data

All data come from Kalshi's public `trade-api/v2` endpoints — no account, no authentication, no order placement. Collection runs in two passes: a universe-wide pass for market metadata, the boundary trades a daily-lookback price panel requires, and closing quotes for the spread filter; then a full-tape pass restricted to contracts surviving the filters, because maker/taker quantities need every fill's price, size and taker side.

The completed tape is **376,760,957 fills across 62 monthly partitions**. The in-scope set for the post-boundary window — markets clearing $1,000 volume, 24 hours open, and a closing bid–ask spread at or under 20c — is **391,427 markets**, all of them taped.

Three construction choices are stated because each silently changes the sample, and because the second is a correction to the natural reading of the prior literature.

Volume is denominated in contracts, not dollars.. The original screens on "We focus only on contracts that have reached a total trading volume upon market closure of at least $1,000", which reads as dollar notional, but Kalshi's `volume` field counts contracts and the API exposes no notional field. At the contract reading our independently collected universe lands within 0.1% of the published event count; the dollar reading roughly halves the sample. Since a sample rule that differentially removes cheap strikes is itself a result in a paper about the favorite–longshot bias, both are computed and the contract reading is primary.

Prices are carried forward on no-trade lookback days.. The original collects "the final traded price as the market closed and also, where available, previous prices from 24-hour intervals up to 10 days before markets closed" — and "where available" reads as skipping empty days. The published counts settle it the other way: 156,986 prices over 46,282 contracts is 3.39 per contract, which skipping cannot reach. On our tape, skipping yields 2.50 and carrying forward yields 3.50. Carrying forward is primary.

One construction on both sides of every boundary.. The estimands below measure a *change* in slope within our own data against our own reference bias. A construction that switched at a boundary is the one thing that would break identification, so the rule is fixed before any estimate is computed.

Categories come from Kalshi's own taxonomy, mapped through a versioned table in the repository.

3. Validation

Before using this pipeline to make a claim about a boundary, we check that it reproduces what is already known about the pre-boundary window. Bürgi et al. (2026), hereafter BDW, provide the reference points.

Table 1: Reproduction of BDW's reference quantities.

Quantity	Ours	BDW
Maker return, price ≥50c	+2.09%	+2.6%
Taker return	−28.40%	−31.46%
Maker share, 1–10c band	43.1%	43.5%
Maker share, 90–99c band	57.0%	56.5%
ψ, 2021	0.0409	0.041
ψ, 2025 (Jan–Apr)	0.0171	0.021

All five of their by-year slopes are recovered with matching sign and significance pattern; magnitudes run slightly below theirs, within roughly one standard error. The bias itself reproduces directly: in the 1–10c band contracts win 2.43% of the time at a mean price of 3.22c, and in the 91–99c band 97.86% at 97.08c.

Our sample is smaller than theirs — 33,222 contracts against 46,282 — and the shortfall is almost entirely the spread filter: 3,257 markets (8.2% of those clearing volume and duration) have no bid–ask history that Kalshi will serve, so the loss is structural rather than a collection gap. The full reconciliation, the construction branches and the divergences are the subject of the companion replication and are not repeated here.

Two checks worth stating. Kalshi's taker-side fields are populated on **100.0%** of all 376.8 million fills in every era. And the two-way (event, contract) cluster-robust variance reduces to one-way event clustering when contracts nest in events (Cameron et al., 2011); estimated both ways on the real panel the standard errors agree to fourteen decimal places, so one-way event clustering is used throughout without hedging.

4. Composition

4.1. The venue changed shape, not just size

Kalshi launched sports contracts in early 2025. The timing is unfortunate for anyone studying the 2025 fee change, because it is essentially the same date.

Table 2: Category weights (shares of in-scope observations), calendar 2024 against the post-boundary window.

Category	Calendar 2024	Post-boundary	Change
Sports	**0.00009**	**0.5875**	+0.5874
Climate & Weather	0.4744	0.0995	−0.3749
Financials	0.3047	0.0059	−0.2989
Mentions	0.0000	0.0646	+0.0646
Economics	0.0690	0.0086	−0.0604

Four categories present after the boundary — Exotics, Mentions, Social, Transportation — did not exist in calendar 2024 at all.

This matters because the slope is category-heterogeneous. Estimated separately, ψ ranges from 0.0107 in Entertainment to 0.0743 in Sports and 0.1206 in Transportation. Re-weighting a heterogeneous quantity is not a neutral act.

4.2. Decomposing the aggregate change

Write the aggregate change in slope at a boundary as

$$\Delta\psi_{\text{agg}} = \underbrace{\sum_c \bar{w}_c\,\Delta\psi_c}_{\text{within}} + \underbrace{\sum_c \Delta w_c\,\bar{\psi}_c}_{\text{between}}, \tag{1}$$

where the first term holds composition fixed at the calendar-2024 mix and varies only within-category behaviour, and the second holds behaviour fixed and varies only weights. At the fee boundary:

The between term is dominated by one category. Sports alone contributes +0.0437 — more than the entire between component, with the shrinking categories contributing negatively against it.

Table 3: Within/between decomposition at the fee boundary.

Term	Value	Share
Within (behaviour)	+0.0399	59%
Between (composition)	+0.0278	**41%**
Aggregate	+0.0677	100%

4.3. What this implies

An aggregate before/after comparison across early 2025 on Kalshi will find a large apparent change in bias, of which roughly two fifths is the arrival of sports contracts. The direction happens to be positive here, which is worth noting: a naive reading of the aggregate would report that the bias *strengthened* after a fee designed to tax it — a conclusion with no plausible mechanism behind it, arrived at by not asking what the sample was made of.

4.4. What is and is not new here

The general principle is old. That an aggregate can move in the opposite direction to every subgroup composing it is Simpson (1951); separating a between-group weight change from a within-group behavioural change is the Oaxaca (1973) and Blinder (1973) decomposition, and the shift-share logic that underlies Bartik (1991) instruments. We claim no novelty for the arithmetic in Section 4.2 and we use the standard form deliberately, so that the object being reported is familiar.

What is specific to this setting is worth stating plainly, because the obvious referee response to Section 4.2 is that it restates a textbook fact.

The magnitude is not a textbook magnitude.. Shift-share decompositions were developed for panels whose composition drifts over decades. A single category going from 0.009% to 58.8% of the sample in fourteen months is not drift; it is the venue becoming a different venue. The resulting distortion is large enough to invert the economic reading rather than merely attenuate it: the naive aggregate says a fee levied on liquidity providers made the market *less* efficiently priced, which is not a mechanism anyone would defend if it were stated aloud.

The original's own robustness check does not reach this.. BDW anticipate the category question and report that "Our results below are not sensitive to cutting the data off in December 2024 or to excluding the category containing sports bets." That check is correct on their sample and does not cover ours, for a reason that is arithmetic rather than methodological: sports is extbf4.3% of the observations inside their window and extbf58.8% after the boundary, a factor of fourteen. Dropping a category worth one observation in twenty-three barely moves an estimate, which is what they found. The problem we identify is created by the share rising to a majority *after* their sample ends, so no test run inside their window could have detected it. Their check also asks a different question — whether a level estimate survives deleting a category — where ours asks how much of a *change across a boundary* is reweighting. Deleting sports and holding the mix fixed are not the same operation.

The trap is live, not hypothetical.. Kalshi is being analysed now, and the samples in circulation straddle exactly this boundary — Becker (2026) covers June 2021 to November 2025, which contains the sports launch in the middle. We are not warning about a possible future error.

And it is a property of young markets specifically.. A mature exchange adds listings within existing asset classes; a new one adds asset classes. Prediction markets are currently doing the latter — Kalshi added sports in 2025, and Polymarket's own composition has moved comparably — so any study of this literature's growth period inherits the problem rather than encountering it occasionally.

The contribution is therefore the measurement and its consequence in a setting where several groups are actively estimating the affected quantity, not the decomposition identity.

The remedy we adopt is to fix the mix at the last full pre-sports year and to report the two terms separately, with every behavioural claim referring to the within term only. Sports is then reported as its own stratum rather than being allowed to dominate an aggregate it did not exist to generate. Any equivalent strategy would do; the point is that some strategy is required, and that its absence is not visible in an aggregate coefficient.

5. What the fee schedule actually says

The design in Section 6 depends on the fee treatment being what we say it is, so we sourced it rather than assuming it. Sixteen dated captures of Kalshi's own published fee document (2021-07 to 2026-02) are archived in the repository with the parse that produces the machine-readable schedule, alongside per-series fee metadata for all 12,231 series obtained from Kalshi's API.

The maker fee was never exchange-wide.. Resting orders are free, in the document's own words, "unless they are included in our 'Maker Fees' section" — an enumerated list of 29 series at introduction, then 39, 48, 101 and 111 by 2025-09-17. On our in-scope universe that is 8.0% of markets and 61.0% of volume in the first-list era, rising to 14.4% and 66.2%. A minority of markets and a majority of dollars — which is why the treatment is economically real and statistically awkward at the same time.

It was flat before it was quadratic.. The first form is round_up($0.0025 \cdot C$) per contract, with no price dependence whatsoever, becoming $0.0175 \cdot C \cdot P(1-P)$ only on 2025-07-08.

It began on 2025-05-13. , by the document's own revision stamp.

Two carve-outs are absent from existing models.. The taker rate was 0.14 in the July 2021 capture, dropping to 0.07 on 2021-08-01. And S&P 500 and Nasdaq-100 markets pay half rate — 0.035 — in every capture from September 2022 onward, covering **18.3%** of the pre-2025 in-scope universe.

One gap is open and carried as bounds rather than closed by assumption: from the 2025-10-01 revision the document stopped enumerating series, so membership after that date is propagated through the fee-sensitivity analysis as evidence bounds (103 series primary, 138 upper).

The methodological consequence is larger than the individual corrections. A step dummy at a single exchange-wide date is the wrong estimator for a treatment that touched a minority of markets, was revised five times, and whose intensity ran 5.5% → 9.4% → 17.6% → 14.5% → 32.0% → 33.5% before decaying. But the same facts supply the remedy: because treatment is per-series and staggered, treated and untreated series trade side by side in the same months.

6. Persistence, properly identified

6.1. The difference-in-differences

We estimate

$$\begin{aligned}(Y-P) = \alpha + \psi P + \alpha_E \text{Ever} + \delta_E(\text{Ever} \cdot P) \\ + \sum_m \left[\alpha_m D_m + \delta_m (D_m \cdot P)\right] + \alpha_D \text{Now} + \delta_D(\text{Now} \cdot P) + \varepsilon, \quad (2)\end{aligned}$$

where Now indicates that the market's series carried a maker fee on the day it closed, Ever marks every observation of a series treated at some point, and D_m are calendar-month dummies entering the slope as well as the level, because the estimand is a slope. δ_D is identified from variation *within a month, between* series that carried the fee and series that did not. Treatment is read from the sourced schedule through the same lookup used for net returns, so the design and the fee accounting cannot disagree about who was treated.

Table 4: The maker-fee difference-in-differences.

Specification	δ_{did}	s.e.	95% CI
Two-way fixed effects	$+0.0113$	0.0129	$[-0.0139, +0.0365]$
Clean controls	-0.0016	**0.0064**	$[-0.0141, +0.0109]$

Estimated on 98 treated series, 141,708 treated observations against 1,188,737 controls, 119,646 event clusters across 60 months, with no cell thin enough to require a wild cluster bootstrap.

Because two-way fixed effects under staggered adoption permits already-treated units to serve as controls for later-treated ones, with the attendant negative weighting (Goodman-Bacon, 2021), both fits are reported always; the clean-controls variant restricts comparisons to never-treated series — the correction Callaway and Sant'Anna (2021) and Sun and Abraham (2021) formalise — and is the one reported on disagreement.

Read on its own, this says the maker fee did not change the average slope. Section 6.3 shows that reading is incomplete.

6.2. The identifying assumption, and why selection sharpens it

δ_{did} is unbiased only under **parallel trends**: absent the fee, treated and untreated series would have moved together in slope. This is an assumption, not a result, and it deserves more scrutiny here than it usually gets, because assignment to treatment was plainly not random.

Kalshi charged an enumerated list of series. Section 5 shows which: 8.0% to 14.4% of in-scope markets carrying 61% to 66% of volume — that is, the series where market-making is most active and most profitable. Maker profitability is not incidental to what we are measuring. It is economically linked to the very slope the estimand concerns: Section 6.6 reports the maker's return advantage as a headline quantity, and an exchange choosing to tax the series where that advantage is largest is selecting on something adjacent to the outcome.

The Ever terms absorb **permanent** differences in level and slope between the charged and uncharged groups, and those differences are certainly present. They do not absorb a **differential trend**. Selection of this kind therefore raises the bar that parallel trends must clear rather than lowering it, and asserting the assumption without evidence would be the weakest point in the paper.

6.3. Event study: pre-trends, and a dynamic effect the average hides

We re-estimate with leads and lags measured in months from **each series' own first treatment month**, never-treated series anchoring the calendar path, $k = -1$ omitted as the reference and endpoints binned. The sample is the same 1,330,445 observations and 119,646 clusters; 98 treated series fall into 11 adoption cohorts.

Table 5: Event-study coefficients around each series' own adoption month.

Event time k	δ_k	s.e.	95% CI
-6	-0.0428	0.0285	$[-0.0988, +0.0131]$
-5	-0.0187	0.0624	$[-0.1410, +0.1037]$
-4	-0.0975	0.0875	$[-0.2690, +0.0740]$
-3	$+0.0016$	0.0633	$[-0.1225, +0.1257]$
-2	-0.0235	0.0410	$[-0.1039, +0.0569]$
$\mathbf{-1}$	—	—	*reference*
0	-0.0092	0.0258	$[-0.0598, +0.0414]$
$+1$	$+0.0179$	0.0278	$[-0.0365, +0.0724]$
$+2$	-0.0120	0.0291	$[-0.0691, +0.0451]$
$\mathbf{+3}$	$\mathbf{-0.0771}$	0.0342	$[\mathbf{-0.1441}, \mathbf{-0.0102}]$
$\mathbf{+4}$	$\mathbf{-0.0826}$	0.0342	$[\mathbf{-0.1496}, \mathbf{-0.0157}]$
$+5$	-0.0462	0.0369	$[-0.1185, +0.0261]$
$+6$	-0.0210	0.0257	$[-0.0713, +0.0293]$
$+7$	-0.0552	0.0346	$[-0.1230, +0.0127]$
$+8$	-0.0242	0.0257	$[-0.0746, +0.0261]$
$+9$	-0.0037	0.0244	$[-0.0515, +0.0441]$

Parallel trends is supported.. Every pre-treatment interval covers zero, and the joint Wald test that all five are zero gives $\chi^2(5) = 3.24$, $p = 0.66$. Slopes were not already diverging before Kalshi charged anybody. Given the selection argument above, this is the evidence the design needed rather than a formality.

The post-treatment coefficients are jointly non-zero — an exploratory finding, labelled as such here and everywhere it is repeated.. The joint test that all ten are zero gives $\chi^2(10) = 21.01$, $p = 0.021$, rejecting at 5%. We report the joint test rather than the two bolded rows, because ten coefficients tested individually at 5% would produce at least one "significant" month roughly forty percent of the time under the null; the joint statistic is what establishes there is something to explain.

The exploratory label is not modesty. This test was not pre-specified and was run after the pre-specified one returned zero — see the status note at the end of this section, which readers should take as attached to every sentence in the next two paragraphs.

The shape is coherent: nothing at impact, a **negative** deviation peaking at three and four months after a series is charged — the favorite–longshot slope *falling* in treated series relative to controls — and decay back toward zero by nine months. So the fee did move the bias, transitorily and with a lag, in the direction of less bias.

We flag two things we cannot settle. The delayed onset is not obviously explicable by a fee that takes effect immediately, and we do not have a mechanism we would defend; candidate explanations include contracts written before treatment settling after it, and adjustment in maker behaviour rather than in posted prices. And the point estimates at $k = +3$ and $+4$ are large relative to the baseline slope of 0.0254 — they imply a temporary sign reversal — with correspondingly wide intervals. We report the shape and the joint test, and we do not build a magnitude claim on two coefficients.

When this analysis was added, and why the two halves differ in status.. The static difference-in-differences was pre-specified in the analysis plan committed on 2026-07-28. The event study was not: it was added on 2026-08-27, after the static estimate was known, in response to a critique of the then-unstated parallel-trends assumption. The two halves of it do not have the same epistemic standing.

The pre-period test is a **validity check on an identifying assumption**. It was worth running whatever it showed, it can only undermine the design rather than manufacture a finding, and we would have reported a failure as readily as this pass. Adding such a check late costs nothing but a date.

The post-period joint test is a **hypothesis test conducted after a pre-specified test returned zero**, and it returned $p = 0.021$. We label it exploratory on that basis. We say so plainly because a transitory effect surfacing in a second analysis, after the first found nothing, is precisely the pattern that innocent specification search produces — and a paper arguing that aggregates hide structure has no standing to be casual about the order in which its own analyses were run.

Accordingly the pre-specified null remains the headline result for the fee question, and the hump is reported as a labelled exploratory finding that warrants a pre-specified test on future data rather than as an established effect.

6.4. What the static estimate does and does not say

The static δ_{did} of -0.0016 (0.0064) is not wrong; it is an average, and its tight interval is a statement about that average rather than about every month within it. Averaging a delayed, transitory, decaying effect over ten post-treatment months is exactly how it becomes indistinguishable from zero.

This is worth naming, because it is the same failure mode as Section 4 in a different guise. There, an aggregate over categories hid the fact that its movement was composition. Here, an aggregate over event time hides the fact that a null is made of a hump. In both cases the aggregate is not false, and in both cases reporting only the aggregate would have been.

The honest summary of the fee question, with each part carrying its status: **pre-specified, no persistent change in the bias; exploratory, a transitory reduction around three to four months after treatment; and separately established, no evidence that the maker's exploitable margin was eliminated** — the last of which Section 6.6 shows directly and which was pre-specified as the fee-sensitivity ribbon.

A limitation we state rather than bury.. With 11 adoption cohorts, the event-study coefficients can still be contaminated across cohorts in the way Sun and Abraham (2021) describe, since a treated cohort's post-period overlaps another's pre-period. Never-treated series carry most of the identifying weight here, which limits the problem but does not eliminate it. A fully interaction-weighted or Callaway–Sant'Anna estimator would settle it, and we regard that as the natural next step rather than something this draft has done.

6.5. *The naive coefficient, and why it disagrees*

The composition-weighted step coefficient at the fee boundary is $+0.0399$ with a 95% interval of $[+0.0071, +0.0726]$ — nominally significant, positive, and therefore implying the bias strengthened under a fee designed to tax it.

Two features explain it and neither is behavioural. First, as Section 5 establishes, the step is applied to a sample most of whose observations were never treated, against a treatment whose intensity quadrupled and then halved inside the window. Second, its within component is essentially one category: Financials contributes $+0.0437$ of the $+0.0399$, on a within-category slope change of $+0.1435$, while holding 30.5% of the composition-held-fixed weight against 0.59% of the current sample. Holding the mix fixed is correct — it is the whole point — but the consequence is that the headline rests on a cell that has shrunk fiftyfold, and we decline to build a persistence claim on it.

6.6. *The exploitable margin survives*

The maker's advantage at prices at or above 50c, in the post-boundary window, computed in three fee layers on 202,725,609 maker and 167,925,545 taker sides with no observation lost to gaps in the fee history:

Table 6: The maker's advantage at prices at or above 50c, by fee layer.

Layer	Margin
Gross	+2.40%
Net of own-era fees	+4.39%
Pre-2025 schedule held constant	+4.54%

The net layer exceeding the gross layer is the expected direction rather than an anomaly: under fees the taker pays and the maker mostly does not, so the spread between them widens. Swept across an eleven-point grid spanning the plausible maker fee band, the margin declines monotonically from $+0.0454$ to $+0.0431$ and **never crosses zero**. There is no break-even rate inside the band.

The margin documented before the boundary is therefore still present after it, and no plausible maker fee erases it.

6.7. *The publication boundary*

The publication coefficient is -0.0311 with an interval of $[-0.0637, +0.0014]$. It contains zero, and it also contains the negative of our reference bias. On this sample, persistence and disappearance cannot be distinguished. We report that rather than choosing.

7. Robustness

Horizon.. The panel pools eleven daily lookback horizons. Estimated separately, the naive fee coefficient swings from $+0.098$ at one day to -0.008 at nine; the one-observation-per-contract closing-price specification gives $+0.0117$ against the pooled $+0.0399$. The difference-in-differences is not subject to this sensitivity, but the failure of the naive coefficient to survive the horizon check is a further reason not to read it behaviourally.

Control venue.. Polymarket's monthly slope over 2025-05 to 2025-12 provides a secular-trend check — not a difference-in-differences, and we do not use parallel-trends language. Three caveats are stated here rather than left for a referee. Polymarket's tail bias runs in the opposite direction to Kalshi's (Qin and Yang, 2026), so the comparison controls for market-wide drift in efficiency and not for level or sign. The Polymarket data come from a public third-party archive, which the author worked with in separate research on Polymarket forecasting — the project this paper's question originated in. That work measures forecasters' skill; this one measures the market's own calibration path, so the archive is put to a different use here, and its full provenance is documented in the repository. And cross-venue arbitrage could propagate the publication event to Polymarket as well, which would strengthen a secular-trend reading rather than undermine the design.

The control window closes at 2025-12-31 for a structural reason: Polymarket began its own staggered fee reform in January 2026, so a longer archive would carry that venue's own treatment inside ours. The fee estimate is unaffected, being identified within Kalshi. **The publication estimate is controlled only through 2025-12-31**, and the remaining six months are reported as an uncontrolled extension of the horizon rather than averaged into a single figure.

8. Related work

Kalshi and the favorite–longshot bias.. Bürgi et al. (2026) is the immediate antecedent and remains a working paper as of this draft. Our companion paper is a full replication of it. The favorite–longshot bias itself is among the most replicated regularities in betting and prediction markets — surveyed by Ottaviani and Sørensen (2008), documented in parimutuel betting by Thaler and Ziemba (1988) and estimated structurally by Snowberg and Wolfers (2010) — and we take its existence as established rather than as something this paper demonstrates. On prediction markets as forecasting instruments generally we follow Wolfers and Zitzewitz (2004).

Whelan (2023) introduces fees into a prediction-market model and shows that fees levied on winnings generate a form of favorite–longshot bias in post-fee loss rates. It is theoretical and uses no Kalshi data. Section 5 speaks directly to its assumptions: the fee it models as uniform is neither uniform nor, in its first eight months, price-dependent.

Becker (2026) analyses 72.1 million Kalshi trades from June 2021 to November 2025, confirms the longshot bias by price level, decomposes returns by market role — takers -1.12% and makers $+1.12\%$ mean excess return per trade — and documents a YES/NO asymmetry in taker order flow. Its role decomposition agrees with ours in direction; magnitudes are not comparable, since the estimand is excess return per trade rather than return on a contract position. It does not address the fee regime, dated boundaries, or composition, and its sample ends seven months before ours. We note that its window straddles the sports launch, which is the configuration Section 4 is about.

Anomalies after publication.. The publication boundary asks a question with an established finance literature behind it. McLean and Pontiff (2016) estimate that cross-sectional return predictors decay by roughly 58% out of sample after publication, of which about a third is attributable to publication itself rather than to in-sample overfitting, and interpret the residual decay as informed trading against a documented signal. The prediction market case is cleaner in one respect and harder in another: contracts settle unambiguously, so there is no benchmark model to misspecify, but a single venue with two candidate treatments four months apart supports far less separation than a cross-section of thousands of predictors. BDW's own closing invitation is this literature's question asked of their result, and we report that our sample cannot answer it.

Staggered difference-in-differences.. Our treatment is adopted at different dates by different series, which is the setting in which conventional two-way fixed effects is now known to misbehave. Goodman-Bacon (2021) shows that the TWFE estimand is a weighted average of all available two-group comparisons, including ones that use already-treated units as controls for later-treated ones, and that those comparisons can receive negative weight when effects are heterogeneous over time. Callaway and Sant'Anna (2021), Sun and Abraham (2021) and de Chaisemartin and d'Haultfœuille (2020) each propose estimators that restrict comparisons to clean control groups and aggregate the resulting group-time effects explicitly.

Our clean-controls specification is the simplest member of that family: it restricts every comparison to treated-versus-never-treated, which is precisely the correction those papers identify as necessary. We report it alongside the TWFE fit rather than instead of it, and treat their agreement as the evidence that weighting is not driving the result. Where they disagree, the clean-controls estimate is the one reported. The event study in Section 6.3 is specified in the same spirit — event time is measured relative to each series' own adoption date, and never-treated series anchor the calendar path.

Composition and decomposition.. The within/between decomposition in Section 4.2 is standard: Simpson (1951) for the aggregation reversal, Oaxaca (1973) and Blinder (1973) for the two-term form, and the shift-share tradition behind Bartik (1991) for holding a mix fixed while letting behaviour vary.

9. Conclusion

A venue that grows by changing what it lists cannot be studied with an aggregate before/after comparison. On Kalshi, 41% of the apparent change in bias at the 2025 fee boundary is the arrival of sports contracts, which went from 0.009% to 58.8% of the sample in fourteen months. The remedy is cheap — fix the mix, decompose, report both terms — and its absence is invisible in the coefficient it distorts.

On the question that motivated the exercise: the maker fee, correctly measured as the per-series and repeatedly revised surcharge it actually was, produced **no persistent change in the bias**. That is the pre-specified result and it stands as the answer.

An event study added afterwards, in response to a referee's question about the identifying assumption, supports that assumption — pre-treatment slopes were not diverging, which the selection into treatment made worth checking rather than assuming — and additionally suggests a transitory reduction peaking three to four months after a series is charged. We label that second finding exploratory and do not treat it as established, because it emerged from an analysis run after the pre-specified one returned zero. If it is real, it should survive a pre-specified test on data this paper does not yet have, and that is the test we would want run before anyone believes it.

The maker's advantage at prices at or above 50c persists after the boundary and survives every plausible fee rate, so whatever the fee did, it did not remove the edge. Whether the public documentation of these patterns changed them cannot be settled on this sample, and we do not claim it can.

The fee history, the category mapping, the archived source documents and the complete pipeline are public.

Declaration of interest

The author declares no competing financial interest. The author does not trade on Kalshi, holds no positions on the venue studied, has no financial or personal relationship with the exchange or with the authors of the work replicated, and received no funding for this research.

Data and code availability

The repository contains the collection pipeline, the analysis code, sixteen archived captures of Kalshi's fee schedule with the parser that turns them into a machine-readable step function, the category mapping table, and the committed analysis plan with its dated amendments. It is MIT-licensed and contains no execution code: every endpoint used is public and unauthenticated.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) in order to draft and revise manuscript prose and to generate the data-collection and analysis code in the accompanying repository. After using this tool the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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